Pakistan SMS classification dataset

Exploratory analysis of 4,895 labelled SMS messages. Source: kaggle.com/datasets/muhammadhumyun/pakistan-sms-classification-dataset

What the dataset is

The file holds 4,895 text messages, each tagged as Normal, Promotional or fraud, plus a pre-lowercased copy of the text. There are no dates, senders or any other fields, and no data dictionary. The stated purpose is training an SMS classifier for Pakistan, so the useful questions are about the text itself: how balanced the labels are, how the classes differ, how much of the material repeats, and whether the messages are real.

The short answer to the last question is: only partly. Everything is in English, with no Urdu script and effectively no Roman Urdu. Reading through it, three kinds of row are mixed together. Some are short, messy, abbreviation-heavy texts with UK premium-rate numbers and prices in pounds, which read like an older English-language spam corpus. Some are Pakistan-specific scam and advertising messages that may well have been collected from real phones. And a large share are clean, polished sentences that come in families of rewordings. Two rows in the file are not messages at all but leftover chatbot replies ("Certainly, here are 50 unique variations of the message...", "And here are 10 more variations"), which settles where those families came from: an LLM was asked to paraphrase seed messages. The source of the older-looking texts is my reading of their style, not something the file documents.

Cleaning

The cleaning was light but it mattered. In the raw file 91.2% of Normal and 70.8% of Promotional messages are wrapped in literal double quotes, against 3.5% of fraud messages. That is a by-product of how the rows were produced, and a classifier could lean on the quote mark alone, so the wrapping quotes were stripped and a flag kept. Label casing was made consistent (fraud became Fraud). The two chatbot rows were removed. Encoding damage was repaired so that a garbled pound sign reads correctly; together with whitespace tidying that touched 77 rows. After normalising, 14 rows were involved in exact duplicates: two messages appeared under both Fraud and Promotional, and since their label is ambiguous all 4 of those rows were dropped, while 5 same-label repeats were reduced to one copy. No rows were removed for length. That leaves 4,884 messages: 2,501 Normal (51.2%), 1,276 Promotional (26.1%) and 1,107 Fraud (22.7%). The file is also sorted by label in three solid blocks, so it has to be shuffled before any modelling.

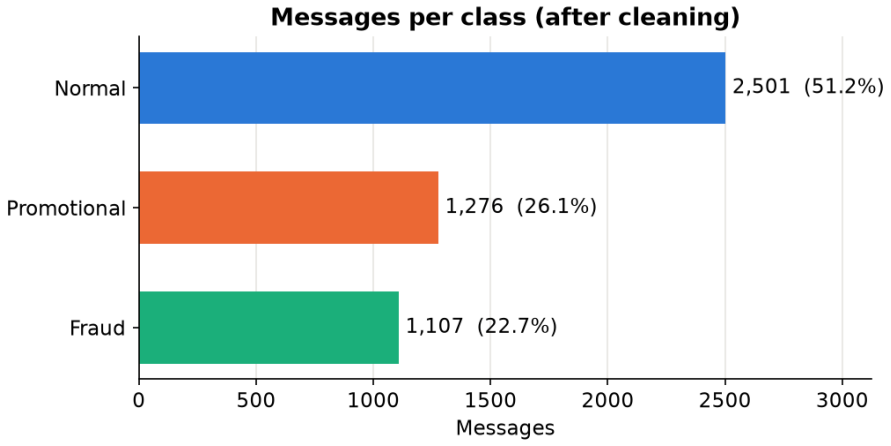


Figure 1. Class sizes after cleaning.

Length

Normal messages are short, with a median of 71 characters and almost none (0.3%) over the 160-character single-SMS limit. Promotional and fraud messages are about twice as long, with medians of 155 and 144 characters, and 43.5% and 27.0% of them respectively run past 160. The shapes differ too. Promotional lengths are spread widely, up to 582 characters, while fraud lengths are packed into a narrow band, which is what you would expect if many of them are variations on a few templates.

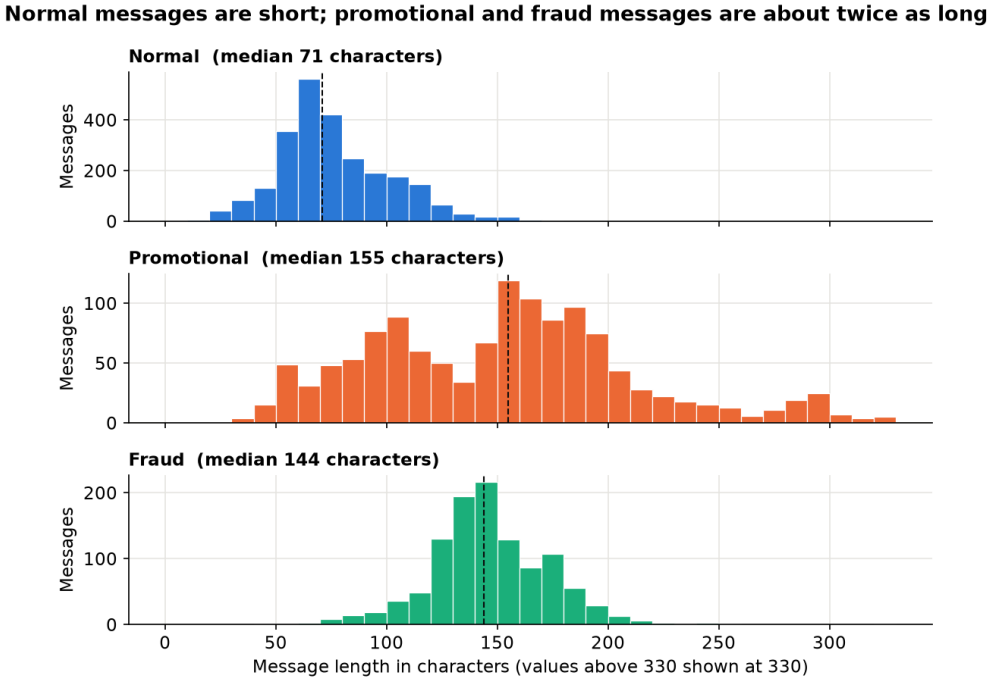


Figure 2. Message length by class. Dashed line marks the median.

Templates and near-duplicates

Exact duplicates are rare, but near-duplicates are not. Swapping every number for a placeholder, 298 fraud messages (26.9%) are word-for-word identical to another fraud message apart from the amount or phone number; the same is true of 4 promotional messages and no normal ones. Using a looser test, TF-IDF cosine similarity of at least 0.8, 40.4% of fraud messages fall into a near-duplicate group, compared with 6.8% of Normal and 4.8% of Promotional. The largest groups have 13 members each and are all the same story: the Benazir Income Support Programme congratulating the reader and awarding a cash prize of some number of rupees. The 0.8 threshold only catches close rewordings, so the true amount of paraphrase is higher than these figures.

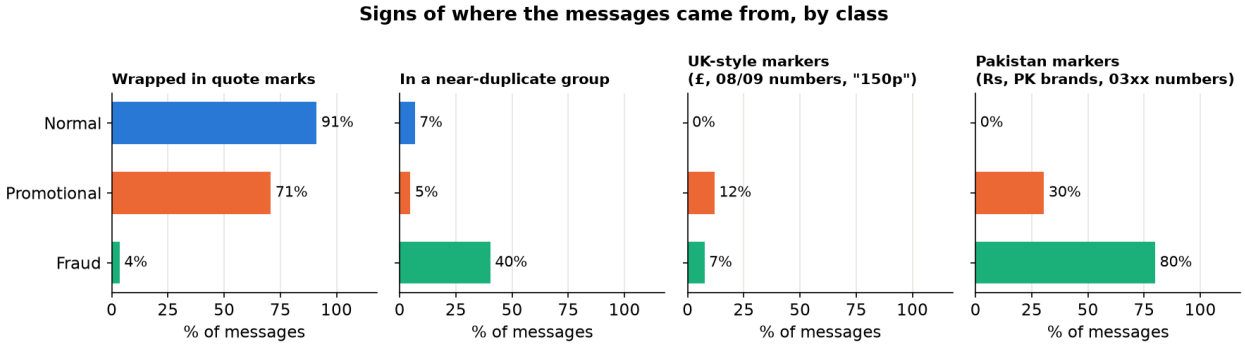


Figure 3. Four rough signals of how each class was assembled. The UK and Pakistan markers are keyword and number-format rules.

Figure 3 puts the provenance signals side by side. The Normal class has no Pakistan markers at all, so it says nothing specific about how people in Pakistan text each other. Fraud is the most Pakistan-specific class (79.9% carry a rupee amount, a local brand or programme, or a local-format mobile number), while 12.0% of Promotional and 7.5% of Fraud messages carry UK-style markers.

What each class contains

The content differences are large. Prize, win or cash wording appears in 72.6% of fraud messages, a money amount in 55.1%, a phone number in 50.0% and the word WhatsApp in 22.4%. Promotional messages look like advertising: 39.7% use discount, offer or free wording, 20.1% carry a hashtag, 18.4% a link and 17.0% a short code or USSD string. Normal messages have essentially none of these; not one contains a phone number, link or hashtag. Urgency wording is less of a fraud marker than one might expect, appearing in 20.8% of promotional messages and 14.4% of fraud ones.

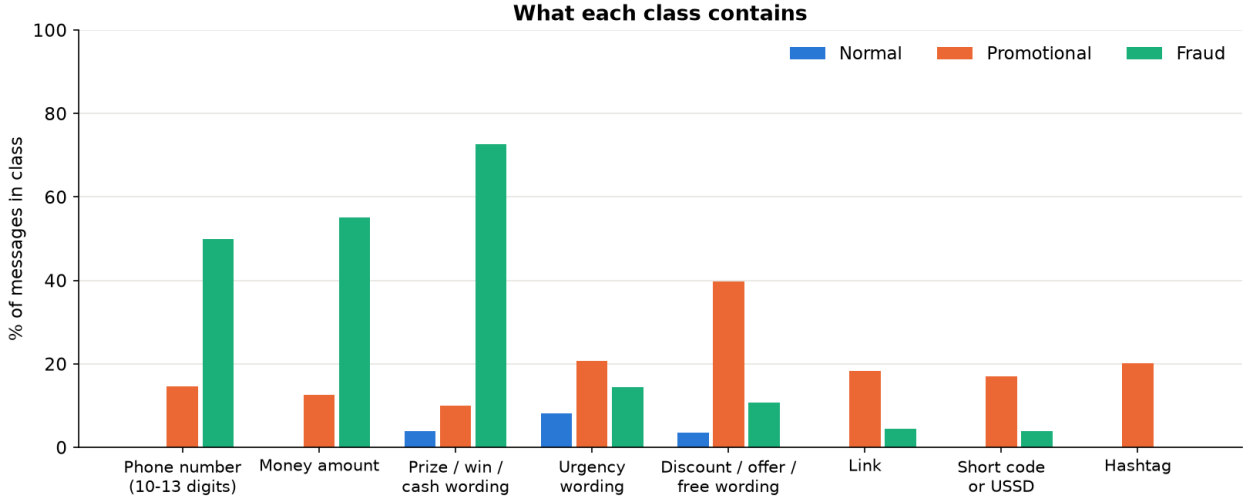


Figure 4. Share of messages in each class containing each pattern (regular-expression flags).

Within fraud, one theme dominates. The Benazir Income Support Programme (or its 8171 code) is mentioned in 48.1% of fraud messages, 527 in total, which makes it the most mentioned name in the whole dataset. Jeeto Pakistan and other TV-show lucky draws account for 17.7%, and bank, ATM or card messages for 15.5%. Telecom and retail brands (Zong, Telenor, Daraz, Easypaisa, Ufone) appear almost only in the promotional class.

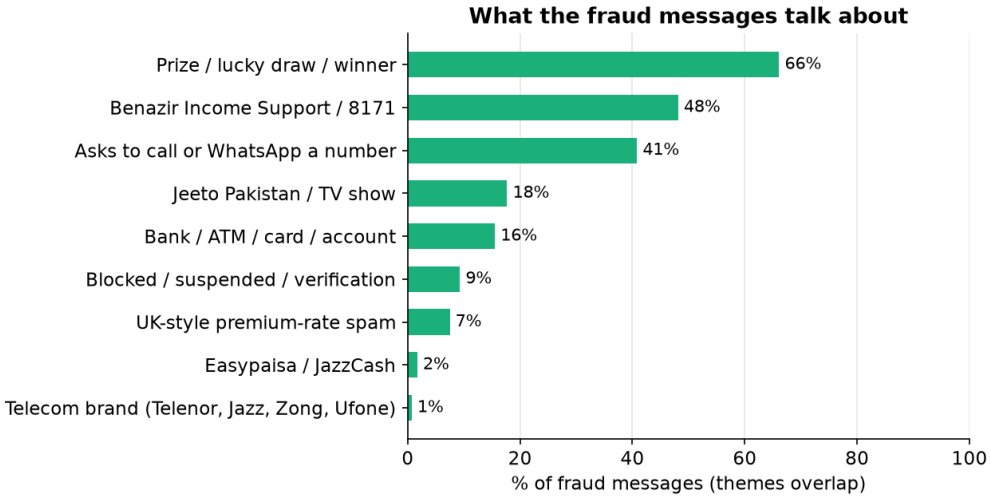


Figure 5. Themes in fraud messages. A message can match more than one.

The phone numbers deserve a note. There are 483 distinct numbers across 742 messages, but two of them alone appear in 51 and 49 fraud messages, and 62 of the distinct numbers contain running-digit sequences of the kind people type as placeholders. This again points to seed messages being reworded with the number kept or invented. Some numbers may still belong to real people, so none are reproduced here. No ID-card-format numbers were found.

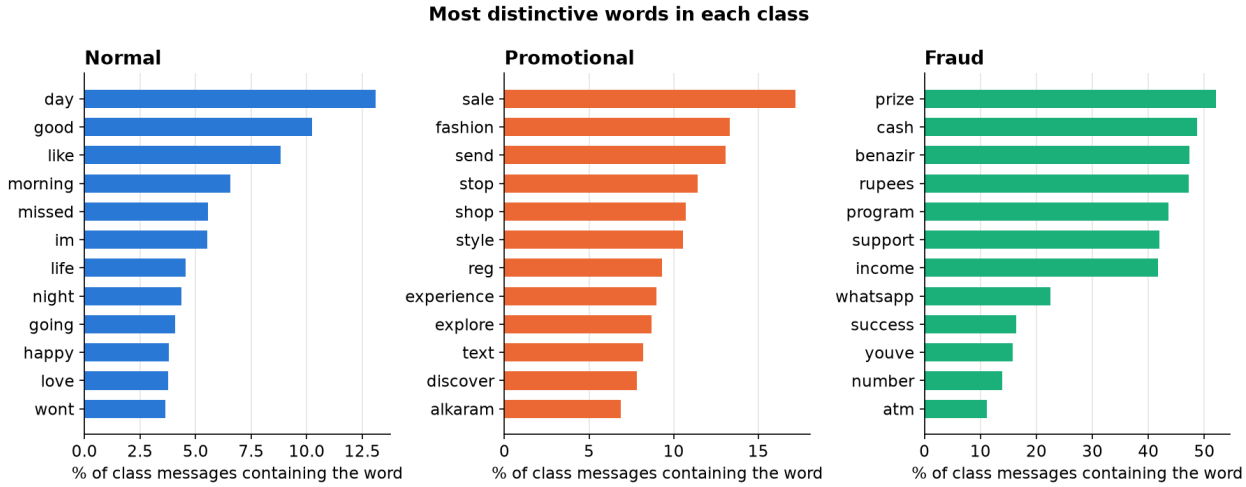


Figure 6. Words that are common in one class and rare in the other two.

How separable are the classes?

As a side check, a plain TF-IDF and logistic regression model was trained on the cleaned text. With an ordinary stratified 80/20 split it scores 0.991 accuracy and 0.988 macro F1, but 12.2% of the test messages have a near-duplicate in the training set. Keeping each near-duplicate group entirely on one side gives 0.983 accuracy and 0.978 macro F1 on 977 test messages, and five-fold versions of both land at about 0.98. So leakage inflates the score only slightly. The classes are simply very different pieces of text: chatty one-liners, brand adverts, and prize scams. Even a three-level decision tree given only the quote-mark flag and the message length reaches 0.81 accuracy on a 0.51 majority baseline.

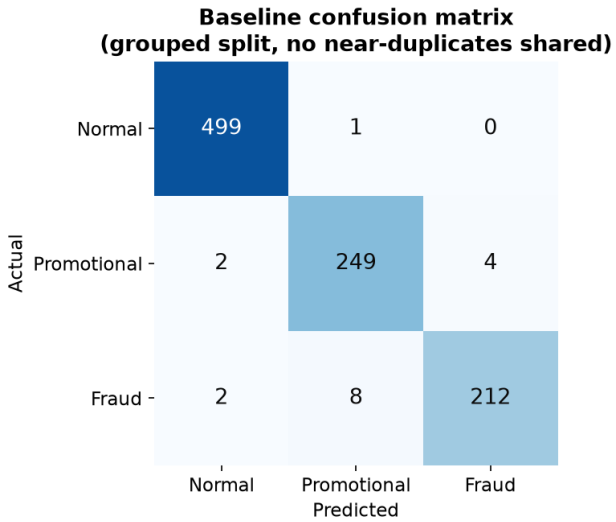


Figure 7. Confusion matrix on the grouped test split (17 errors out of 977).

Of the 17 errors, 12 are confusions between Fraud and Promotional, and most of those are UK-style spam that is filed under Fraud in some rows and Promotional in others. The boundary between those two labels is not applied consistently for that material.

Highest, lowest, most

- Largest class is Normal with 2,501 messages (51.2%); smallest is Fraud with 1,107 (22.7%).
- Longest message is 582 characters (Promotional); shortest is 12 characters (Normal).
- Median length: Normal 71, Fraud 144, Promotional 155 characters.
- Most templated class: Fraud, with 40.4% of messages in a near-duplicate group, against 6.8% and 4.8%.
- Largest near-duplicate group: 13 messages, all Benazir cash-prize rewordings.
- Most common fraud theme: Benazir Income Support, in 48.1% of fraud messages.
- Most reused phone number: 51 fraud messages share one number; the runner-up appears in 49.
- Largest plain rupee figure quoted in a fraud message is 84,500; the median is 23,500 (457 messages).
- Hashtags appear in 20.1% of promotional messages and in no fraud or normal message.
- Quote-wrapped in the raw file: 91.2% of Normal, 70.8% of Promotional, 3.5% of Fraud.

Conclusion and limits

This is a usable toy dataset for a three-way text classifier, and it does contain a genuine catalogue of scam storylines aimed at Pakistani phone users. It should not be read as a sample of real SMS traffic. It is English only, a good part of it is machine-written paraphrase, part of it appears to be borrowed from an older foreign corpus, and the three classes were clearly assembled in different ways. The near-perfect classifier scores describe this file and nothing more.

Other limits: the provenance calls are judgements from the text and its artefacts; the near-duplicate grouping depends on a 0.8 similarity cut-off and misses looser rewordings; the pattern flags are regular expressions with some false hits and misses; amounts written in words such as "5 lakh" are not in the rupee figures; and with no timestamps or senders nothing can be said about timing or campaigns.

Worth trying next: testing a model trained here against independently collected Urdu and Roman Urdu messages; clustering the fraud class into storylines at a looser threshold to count how many distinct seed messages there really are; and re-labelling the UK-style spam consistently before using the Fraud versus Promotional boundary for anything.